AI-Powered Observability & API Failure Prediction Platform

System Architecture

Client Apps → Metrics SDK → Ingestion API → Queue (Redis) → Processing Worker → MongoDB → AI Engine → Dashboard (React + WebSockets)

Features

1. Metrics Collection (SDK): Collect latency, status code, endpoint data. 2. Ingestion API: Receive metrics and push to Redis queue. 3. Queue System (Redis): Buffer incoming data and handle load. 4. Processing Worker: Process, clean, and store data. 5. Database (MongoDB): Store raw and aggregated metrics. 6. Dashboard (React): Visualize latency, errors, and request count. 7. Real-Time Updates (Socket.IO): Live updates without refresh. 8. Anomaly Detection: Detect unusual patterns using Z-score and moving average. 9. Alert System: Trigger alerts on anomalies or thresholds. 10. Failure Prediction: Predict possible API failures using trend analysis. 11. Chaos Testing: Simulate failures and delays. 12. Root Cause Indicator: Identify problematic services.